

Chapter 6

Probability

6.1 Basic ideas

In measuring the uncertainties of everyday life, ambiguous terminology, such as, "fairly certain" and "highly likely" is used. Mathematics attempts to assign a number to the likelihood of the occurrence of an event. This measurement is called the **PROBABILITY** that the event will occur.

For example, when a coin is tossed, the probability that it will land heads up is $\frac{1}{2}$. Any happening whose result is uncertain is called an **experiment**. The possible results of an experiment are called **outcomes**. The set of all possible outcomes of an experiment is the sample space(S), and any subcollection of a sample space is an **event** (E).

Example 6.1.1. Find the sample space for each of the following experiments:

1. One coin is tossed.
2. Two coins are tossed.

Solutions

1.

$$S = \{H, T\}.$$

2.

$$S = \{HH, TT, HT, TH\}.$$

The sample space

$$S = \{HH, TT, HT, TH\}$$

distinguishes between the two cases HT and TH , even though the two outcomes appear to be similar. Another way to list possible outcomes is using a tree diagram, as shown below

When an event E has equally likely outcomes and its sample space S has equally likely outcomes, the probability of an event E is

$$P(E) = \frac{\text{Number of outcomes in event } E}{\text{Number of outcomes in the sample space } S}.$$

More formally, the number of outcomes in event E is denoted by $n(E)$, and the number of outcomes in the sample space S is denoted by $n(S)$.

6.2 Probability of an event

If an event E has $n(E)$ equally likely outcomes and its sample space S has $n(S)$ equally likely outcomes, then the probability of an event E is

$$P(E) = \frac{n(E)}{n(S)}.$$

6.2.1 Properties of the probability of an event

Let E be an event that is a subset of a finite sample space S .

1. $0 \leq P(E) \leq 1$.
2. If $P(E) = 0$, then E cannot occur and is called an impossible outcome. It is impossible that you will be 20 years on your next birthday if you 15 years today.
3. If $P(E) = 1$, then E must occur and is called a certain outcome. If today is Monday, then it is certain that tomorrow is Tuesday.

Example 6.2.2. 1. Two coins are tossed. What is the probability that both land heads up?

2. A card is drawn at random from a deck of playing cards. What is the probability that it is an ace.

Example 6.2.3. Two six sided dice are tossed. What is the probability that the total of the two dice is 7?

6.3 Mutually Exclusive events

Two events A and B , from the same sample space, are said to be **mutually exclusive** when A and B have no outcomes in common. Consequently,

$$P(A \cap B) = 0.$$

Examples are: Drawing a red card and drawing a club cannot occur at the same time, because all clubs are black; therefore, the outcomes drawing a red card and drawing a club are mutually exclusive. Running forward and backwards at the same time. Tossing a coin.

6.3.1 Rules of Probability

1. Complement events. If the probability of an event E happening is $P(E)$, then the probability that it does not happen is $1 - P(E)$, since

$$\sum_{i=1}^n P(E_i) = 2.$$

This is called the complement of E , E^c , and so

$$P(E^c) = 1 - P(E).$$

2. Union of two events-Addition rule.

The probability of either two events E_1 and E_2 occurring is given by

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2),$$

or

$$P(E_1 \cup E_2) = P(E_1) + P(E_2),$$

if E_1 and E_2 are mutually exclusive.

Example 6.3.2. Two dice are thrown. What is the probability of scoring either a double, or a sum greater than 8.

Solution Let E_1 be an event of scoring double and E_2 be the event of scoring a sum greater than 8. It follows that

$$E_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

and

$$E_2 = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

$n(S) = 36$, $n(E_1) = 6$, $n(E_1 \cap E_2) = 2$ and $n(E_2) = 10$. Hence,

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2) = \frac{6}{36} + \frac{10}{36} - \frac{2}{36} = \frac{7}{18}.$$

3. Intersection of Two Events- Conditional Probability

If E_1 and E_2 are two events, then the probability that E_2 occurs given that E_1 has occurred is the probability of E_2 given E_1 and is defined as

$$P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)}$$

Example 6.3.3. Two dice are thrown. What is the probability of scoring a double given that the total on the two dice is greater than 8?

Solution Let E_1 be an event of scoring double and E_2 be the event of scoring a sum greater than 8 as in the previous example. Then

$$E_1 = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6)\}$$

and

$$E_2 = \{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}.$$

Consequently,

$$P(E_1 \cap E_2) = \frac{2}{36}, \quad P(E_2) = \frac{10}{36}.$$

Therefore,

$$P(E_1/E_2) = \frac{P(E_1 \cap E_2)}{P(E_2)} = \frac{\frac{2}{36}}{\frac{10}{36}} = \frac{1}{5}.$$

Example 6.3.4. It is given that two computers in a batch of 6 computers are faulty, but it is not known which two. If two computers are selected at random, one at a time without replacement. Find the probability that they are both faulty.

Solution

Let E_1 be the event that the first computer is faulty, and E_2 be the event that the second computer is faulty, we seek to find $P(E_1 \cap E_2)$. From

$$P(E_2/E_1) = \frac{P(E_2 \cap E_1)}{P(E_1)},$$

we get

$$P(E_2/E_1)P(E_1) = P(E_2 \cap E_1).$$

But

$$P(E_1) = \frac{1}{3}$$

and

$$P(E_2/E_1) = \frac{1}{5}.$$

Hence

$$\frac{1}{5} \times \frac{1}{3} = P(E_2 \cap E_1) = \frac{1}{15}.$$

6.4 Independent Events

Two events A and B , from the same sample space, are said to be **independent** if and only if

$$P(A \cap B) = P(A)P(B).$$

This result can be extended to any number of events.

Example 6.4.1. 1. If the probability that person A will be alive in 20 years is 0.7 and the probability that person B will be alive in 20 years is 0.5, what is the probability that they will both be alive in 20 years?

2. A fair die is tossed twice. Find the probability of getting a 4 or 5 on the first toss and a 1, 2, or 3 in the second toss.

6.5 Bayes' Theorem

Before we state the Bayes' Theorem, we have

Theorem 6.5.1. *Let $A_1, A_2, \dots, A_n$ form partition of the sample space with $P(A_i) \neq 0$ for all i , and let B be any event. Then*

$$P(B) = \sum_{i=1}^n P(B/A_i) \cdot P(A_i).$$

Theorem 6.5.2. *Let A and B be events with non-zero probability. Then*

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B)},$$

which follows from using the conditional probability twice.

If $P(A) \neq 0, 1$ and $P(B) \neq 0$, then Bayes' Theorem becomes

$$P(A/B) = \frac{P(B/A) \cdot P(A)}{P(B/A) \cdot P(A) + P(B/A^c) \cdot P(A^c)}.$$

We always state the Bayes' Theorem in this form.

Example 6.5.3. In a survey, 70% of men and 55% of women said that they smoked. The Proportion of men to women was 60 : 40 and a person is chosen at random and found to be a smoker. What is the probability that this person is a woman?

Solution Let E_m be the event " a man is chosen"

Let E_w be the event " a woman is chosen"

Let E_s be the event " a person is a smoker"

We seek to find

$$P(E_w/E_s).$$

Now,

$$P(E_w/E_s) = \frac{P(E_w \cap E_s)}{P(E_s)},$$

so that by Bayes' Theorem

$$P(E_w/E_s) = \frac{P(E_s/E_w) \cdot P(E_w)}{P(E_s/E_w)P(E_w) + P(E_s/E_m)P(E_m)}.$$

But $P(E_s/E_w) \cdot P(E_w) = \frac{11}{20} \cdot \frac{2}{5} = \frac{11}{50}$, and

$$P(E_s/E_w)P(E_w) + P(E_s/E_m)P(E_m) = \frac{11}{20} \cdot \frac{2}{5} + \frac{7}{10} \cdot \frac{3}{5} = \frac{32}{50}.$$

Consequently,

$$P(E_w/E_s) = \frac{P(E_s/E_w) \cdot P(E_w)}{P(E_s/E_w)P(E_w) + P(E_s/E_m)P(E_m)} = \frac{\frac{11}{50}}{\frac{32}{50}} = \frac{11}{32}.$$

Such problems involve finding the probability of a second event happening knowing that a first event has occurred.

If it is not easy to understand the solution in the previous example, it may be easier to find the solution by using a tree diagram as **shown below**.